Your grade: 100%

Your latest: **100%** • Your highest: **100%** • To pass you need at least 80%. We keep your latest score.

Next item →

Instructions

1. A cybersecurity team is working to implement an anomaly detection system to monitor network traffic for signs of botnet activity. They are aware of several potential challenges they might face.
Which of the following challenges are they likely to encounter in this scenario? (Select all that apply)

1 / 1 point

- [x] Scalability issues

  Nice work

  Correct! As the volume of botnet data increases, scalability becomes a significant challenge for the detection system.

- [ ] Data encryption overhead
- [x] High false positive rate

  Nice work

  Correct! The high false positive rate can overwhelm the team with alerts, making it difficult to identify genuine threats.

- [ ] Lack of historical data
- [x] Evolving attack patterns

  Nice work

  Correct! Attackers' continuous evolution of methods can complicate the task of updating and maintaining effective anomaly detection models.

2. Which strategies are effective in detecting anomalies in cybersecurity, particularly with botnet data? (Select all that apply)

1 / 1 point

- [x] Signature-based detection

  Nice work

  Correct! Combining signature-based detection with anomaly detection can enhance overall security by recognizing known threats.

- [ ] Manual log inspection
- [x] Behavioral analysis

  Nice work

  Correct! Behavioral analysis is essential for identifying deviations from typical patterns of activity.

- [x] Heuristic analysis

  Nice work

  Correct! Heuristic analysis can uncover suspicious patterns by evaluating deviations from expected behavior.

- [ ] Static IP addressing

3. Which machine learning technique is frequently utilized for detecting anomalies in network traffic?

1 / 1 point

- ( ) K-means Clustering
- ( ) Linear Regression
- ( ) Convolutional Neural Networks (CNN)
- (•) Support Vector Machines (SVM)

  Nice work

  Correct! Support Vector Machines are well-regarded for their ability to handle high-dimensional data and identify anomalies effectively.

4. What is a fundamental component of an effective anomaly detection system?

1 / 1 point

- (•) Feature extraction
- ( ) Data encryption
- ( ) User authentication
- ( ) Network routing

  Nice work

  Correct! Feature extraction is crucial for transforming raw data into a format that reveals patterns and anomalies.

5. In the context of botnet activities, what role does a 'command and control' server play?

1 / 1 point

- ( ) A server that stores backup data
- (•) A server that sends commands to compromised devices
- ( ) A server that routes IP packets
- ( ) A server that encrypts network traffic

  Nice work

  Correct! Command and control servers are integral to botnet operations, directing the activities of infected devices.